

Safety-Critical Multi-Sensor Acquisition Framework with Priority-Deadline Hybrid Scheduling and Hybrid Memory Optimization on STM32

Pemrograman Kontroler — Kelompok 10/4B

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Latar Belakang & Permasalahan



Sistem tertanam untuk instrumentasi industri harus menggabungkan: akuisisi multi-sensor, jaminan real-time, efisiensi energi, dan keselamatan tingkat tinggi semua pada MCU berresource terbatas. Studi Scopus/WoS menawarkan solusi parsial (adaptive sampling, TCM/CCM, deadline-aware scheduling, CBF) tetapi jarang diintegrasikan dan diuji fungsional dalam satu kerangka kerja.

Proyek ini menutup gap tersebut dengan menyintesis 16 future-work menjadi satu framework yang dapat diuji di Proteus pada STM32F401VE (simulasi).

Tujuan & Ruang Lingkup

Tujuan

1. Dokumentasikan 16 jurnal & future work
2. Rancang metode sintesis
3. Implementasi & uji fungsional (T1–T10) di Proteus
Bandingkan dengan metode konvensional.

Ruang Lingkup

Simulasi: STM32F401VE @84 MHz (menggantikan F407VGTx). Perbedaan utama: tidak ada CCM SRAM, clock lebih rendah. Artefak: kode, skema Proteus, CSV, skrip analisis.

Repository: github.com/AbigailNovariant/stm32-adaptive-safety-framework

Studi Literatur — Ringkasan 16 Jurnal

Tabel 1: Ringkasan 16 jurnal internasional (Scopus/WoS, 2021–2026)								
No	Judul (ringkas)	Penulis	Thn	Jurnal	Topik	Metode	Hasil Utama	Future Work
1	Comparison of Embedded System Design for Industrial Applications	A. Malinowski, H. Yu	2011	IEEE TII	#1	Survei MCU-/FPGA/DS-P/ASIC	Peta platform industri	FW1: Platform hibrid FPGA+MCU <i>safety-critical</i>
2	Energy-Efficient Dual-Core RISC-V Architecture for Edge AI	C. A. Tanase	2026	Computers (MDPI)	#1	MAC sharing + DFS	Efisiensi energi inferensi AI	FW2: Multi-MAC arbitrase hierarkis
3	Hardware-Accelerated Digital Power Control for High-Frequency Hybrid Energy Storage	W. Lin, G. Chu	2025	IET Power Electr.	#1	FMAC kompensator 250 kHz	MCU umum menggantikan DSP	FW3: Timing ADC / mitigasi <i>sampling delay</i>
4	Quantization and Deployment of Deep Neural Networks on Embedded Targets	P.-E. Novac et al.	2021	Sensors (MDPI)	#1	Post-training & QAT	DNN di MCU dengan trade-off daya	FW4: Optimasi <i>memory footprint</i> DNN
Tabel 1 — lanjutan								
No	Judul	Penulis	Thn	Jurnal	#	Metode	Hasil	FW
5	RTOS-Integrated Time Synchronization for Self-Deployable Wireless Sensor Networks	S. Goossens et al.	2026	Sensors (MDPI)	#2	UTC GNSS RTOS	via Sinkronisasi di presisi WSN	FW5: Frekuensi sync adaptif; timestamp di paket data
6	Ultralight Situation-Aware Hypervisor With Adaptive Starvation-Free Feedback Control	M. Kim, D. Park	2025	IEEE Access	#2	RR + prioritas + feedback	Multi-OS overhead $O(N)$	FW6: <i>Deadline-based + priority-based</i>
7	uTango: An Open-Source TEE for IoT Devices	D. Oliveira et al.	2022	IEEE Access	#2	TrustZone-M TEE	Isolasi aplikasi IoT	FW7: Secure storage, kriptografi, OTA
8	Design and Implementation of an Atrial Fibrillation Detection Algorithm on the ARM Cortex-M4	M. Zylinski et al.	2023	Sensors (MDPI)	#3	Pipeline DSP+ML wearable	ML lengkap di M4	FW8: Multi-aritmetika, latensi rendah

No	Judul	Penulis	Thn	Jurnal	#	Metode	Hasil	FW
9	LLVM-Based Efficient Hybrid Cache and TCM Memory Allocation for Low-Latency	G. Jeon, D. Park	2025	IEEE ESL	#3	Compiler pass Cortex-M7	Latensi turun tanpa HW baru	FW9: Multi-core, pola akses dinamis
10	An Ultra-Low-Power Time-Domain Level-Crossing ADC With Adaptive Sampling Rate	N. Jiang et al.	2025	IEEE TCSII	#4	<i>Level-crossing</i> non-uniform	Daya ADC turun drastis	FW10: CMOS maju, biomedis BW tinggi
11	Front-End Gateway System With Serial Communication Protocol Conversion and Edge Computing	G.-M. Sung et al.	2023	IEEE Access	#4	Konversi protokol + edge	Bus dan latensi berkurang	FW11: SPI/CAN + keamanan protokol

No	Judul	Penulis	Thn	Jurnal	#	Metode	Hasil	FW
12	A C-Based Framework for Low-Cost Real-Time Embedded Systems	I. C. Bertolotti	2025	Future Internet	#5	Abstraksi multi-jaringan	Overhead WCET <6%	FW12: Protokol baru, keamanan, multi-RTOS
13	Energy Efficient Mixed Task Handling on Real-Time Embedded Systems Using FreeRTOS	D. Ramegowda, M. Lin	2022	J. Syst. Archit.	#5	Mixed task DVFS	Hemat energi task campuran	FW13: Multi-processor di kernel FreeRTOS
14	Implementation of a Universal Framework Using SOLID Design Patterns for Microcontrollers	M. Babusch, P. Folynek	2024	Sensors (MDPI)	#5	Design pattern firmware	<i>Maintainability</i> meningkat	FW14: MCU terbatas, CI/CD

No	Judul	Penulis	Thn	Jurnal	#	Metode	Hasil	FW
15	Embedded Control Barrier Functions for Safety-Critical Control of High-Relative-Degree Systems	Z. Li et al.	2025	IEEE TII	#6	EMB-CBF-SCC nonlinier	Constraint safety terjaga	FW15: Ketidakpastian tinggi, MCU industri
16	Security and Functional Safety for AI in Embedded Automotive Systems: A Tutorial	Y. Wang et al.	2024	IEEE TCSII	#6	ISO 26262 + adversarial	Peta jalan safety AI	FW16: Standar AI, verifikasi formal ML

Dari Future Work (FW1–FW16) ke Komponen Sistem

Tabel 2: Pemetaan FW ke komponen implementasi

Komponen Implementasi	Sumber FW
ADC DMA + <i>adaptive sampling rate</i>	FW3, FW10
Buffer kritis (CCM / SRAM DMA prioritas tinggi)	FW9
Penjadwalan hibrid FreeRTOS (4 task, deadline-aware)	FW6, FW13
Gateway multi-protokol UART + I2C	FW11
CBF threshold + WWDG + FSM (NORMAL–WARNING–FAULT)	FW15, FW16
Firmware modular (SOLID-ready, CI/CD-compatible)	FW12, FW14

Tabel 4: Konfigurasi alat simulasi

Komponen	Detail
MCU	STM32F401VE, Cortex-M4 @ 84 MHz
RTOS	FreeRTOS (CMSIS-RTOS v1)
IDE	Keil μ Vision 5 + STM32CubeMX
Simulator	Proteus 8.x
Sensor analog	POT1 (PA0), POT2 (PA1), range 0–3,3 V
PWM output	TIM1_CH1 (PA8), frekuensi 10 kHz
Komunikasi	USART1 (115200 bps) + I2C1 (PCF8574)
Indikator	LED PD12 (Normal), PD13 (Warning), PD14 (Fault)
Ambang CBF	$V_{\min} = 200\text{ mV}$, $V_{\max} = 2800\text{ mV}$

Empat Pilar Arsitektur

1

1. Sampling Adaptif (FW3, FW10)

Pembacaan sensor menyesuaikan keadaan. Jika nilai sensor berubah drastis, mikrokontroler akan membaca dengan cepat (tiap 10ms). Jika nilai sensor stabil, ia membaca lebih lambat (tiap 100ms) untuk menghemat daya.

2

2. Memori Hibrid (FW9)

Menggunakan teknik DMA (Direct Memory Access) agar perpindahan data sensor tidak membebani prosesor utama.

3

3. Penjadwalan Hibrid (FW6, FW13)

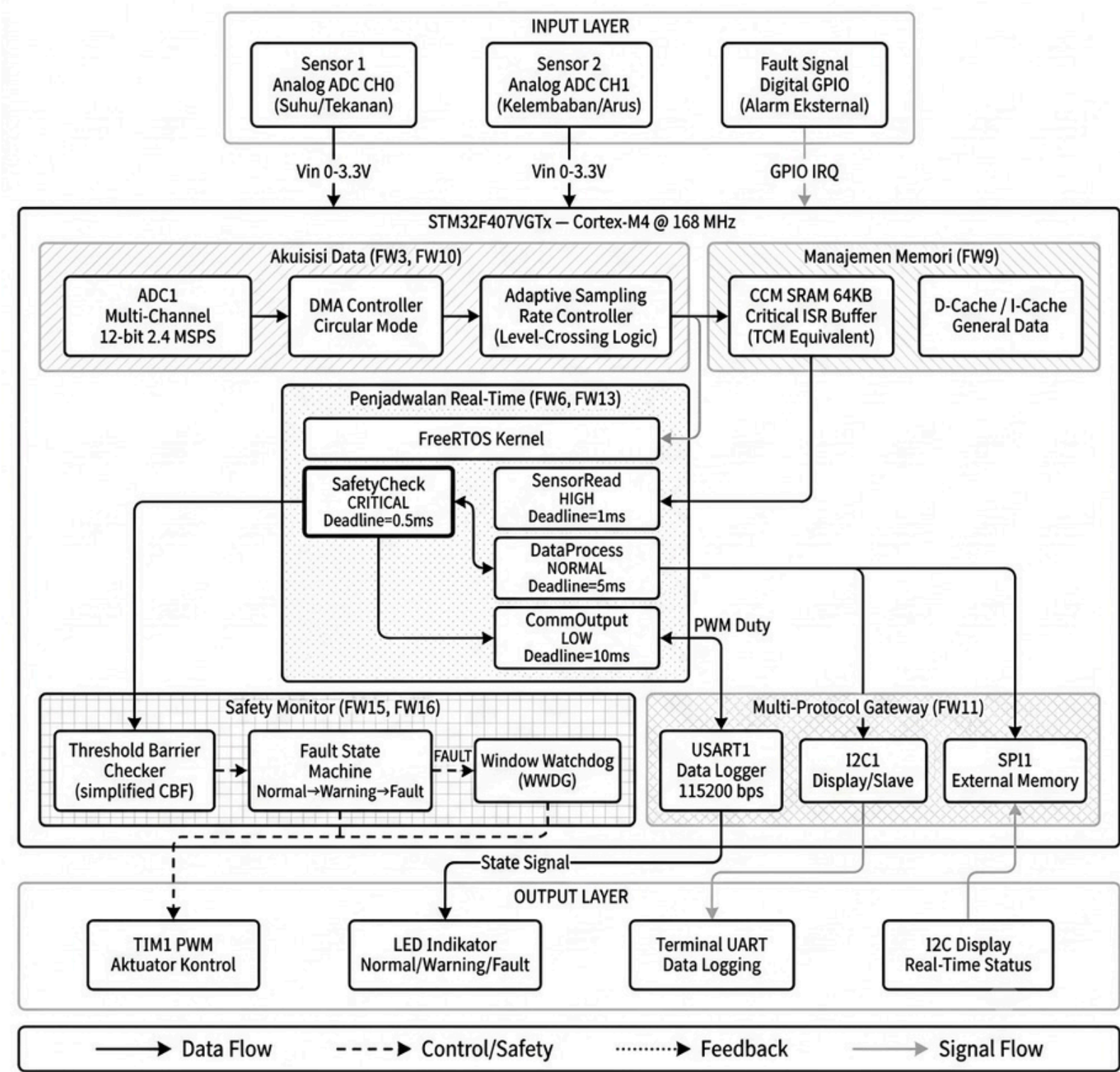
Membagi tugas prosesor menjadi 4 prioritas. Tugas "Keamanan/Safety" adalah yang paling penting dan akan didahulukan daripada tugas mencetak teks ke layar (Komunikasi).

4

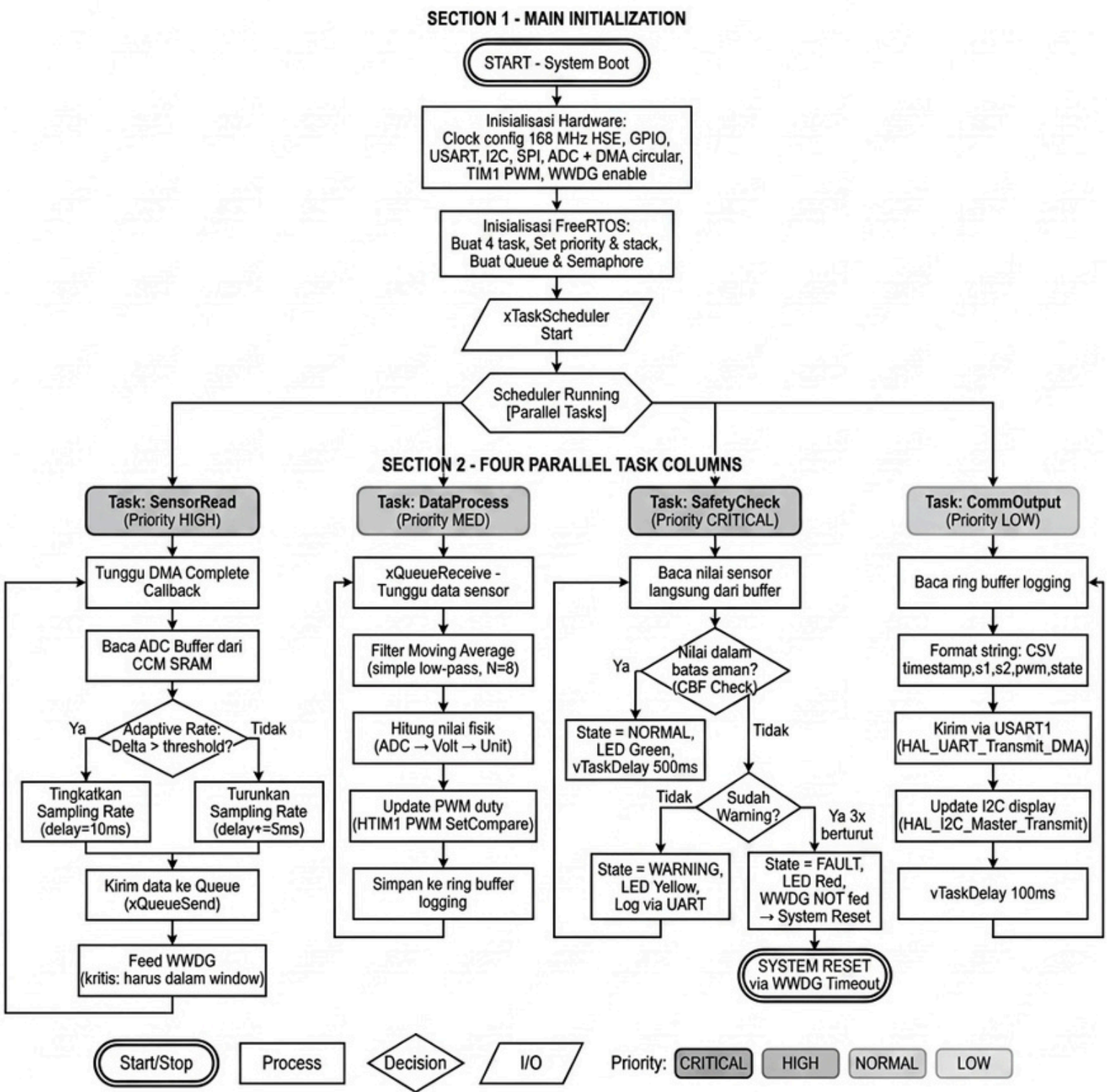
4. Safety Barrier Berlapis (FW15, FW16)

Sistem punya batas aman sensor (200 mV - 2800 mV). Jika lewat batas 1 kali, lampu WARNING nyala. Jika 3 kali berturut-turut lewat batas, status jadi FAULT, dan sistem akan mengaktifkan Watchdog (WWDG) untuk me-restart/mematikan sistem.

Diagram Blok & Flowchart Sistem



Gambar 1. Diagram blok arsitektur framework



Gambar 2. Flowchart program utama - Inisialisasi sistem dan 4 task paralel FreeRTOS

Rangkaian Proteus

MCU

STM32F401VE,
Cortex-M4 @84
MHz

RTOS / IDE

FreeRTOS
(CMSIS-RTOS v1) ·
Keil µVision5 +
CubeMX

Simulator

Proteus 8.x

Perangkat

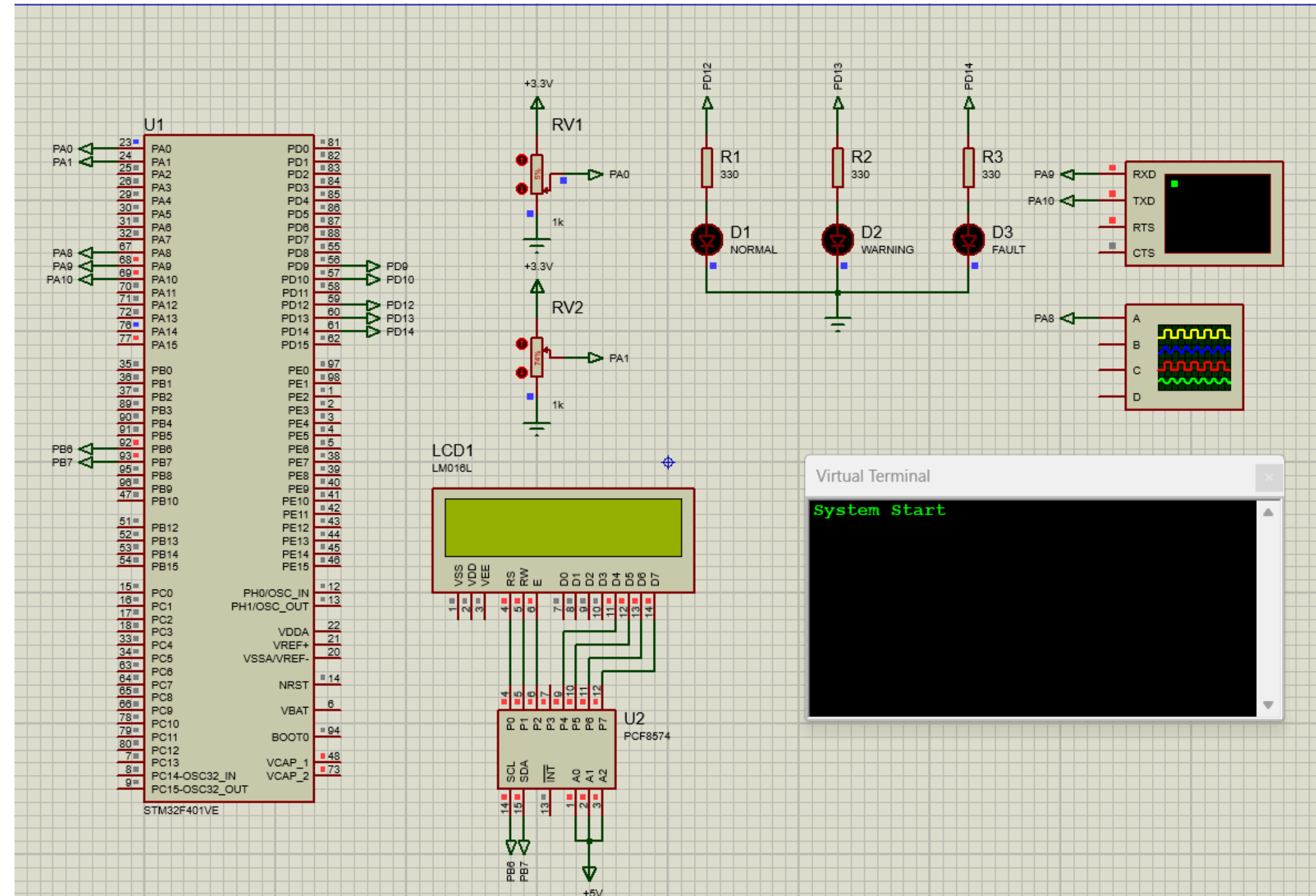
POT1/PA0,
POT2/PA1, TIM1
CH1 (PA8) PWM 10
kHz, USART1
115200, I2C1
PCF8574

Indikator

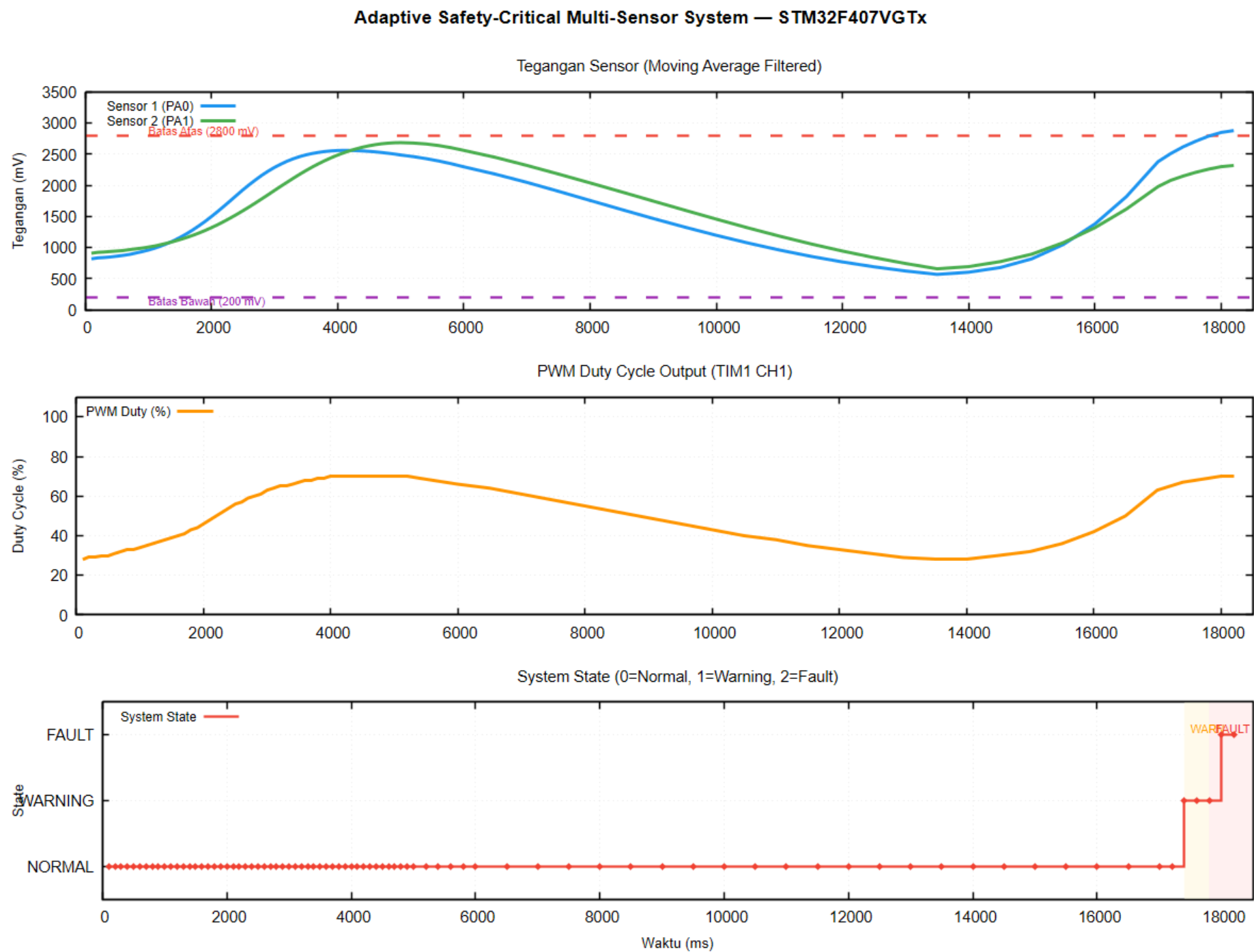
LED PD12 (Normal),
PD13 (Warning),
PD14 (Fault)

CBF Ambang

Vmin=200 mV,
Vmax=2800 mV



Rancangan Pengujian Fungsional (T1–T10)

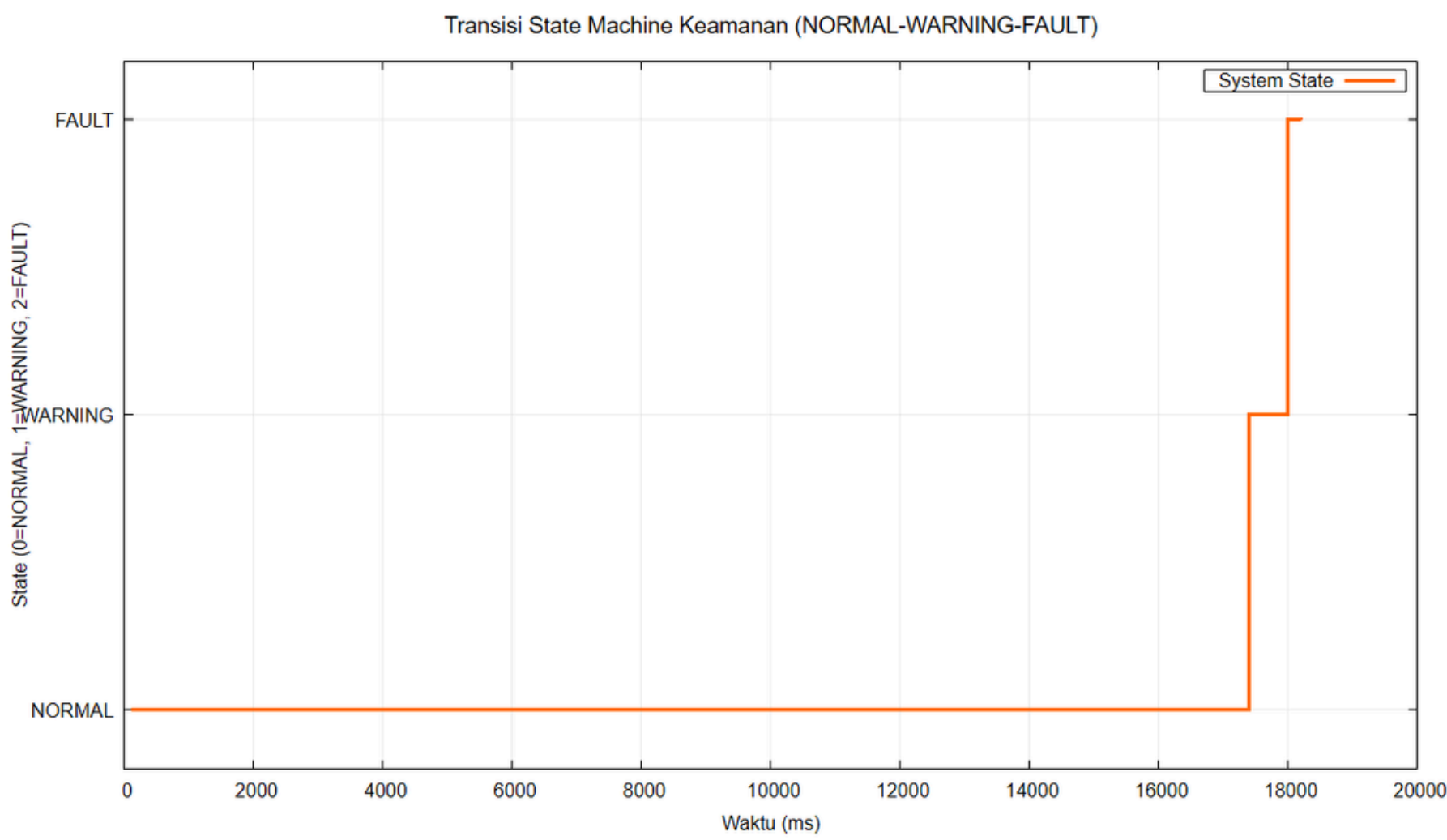
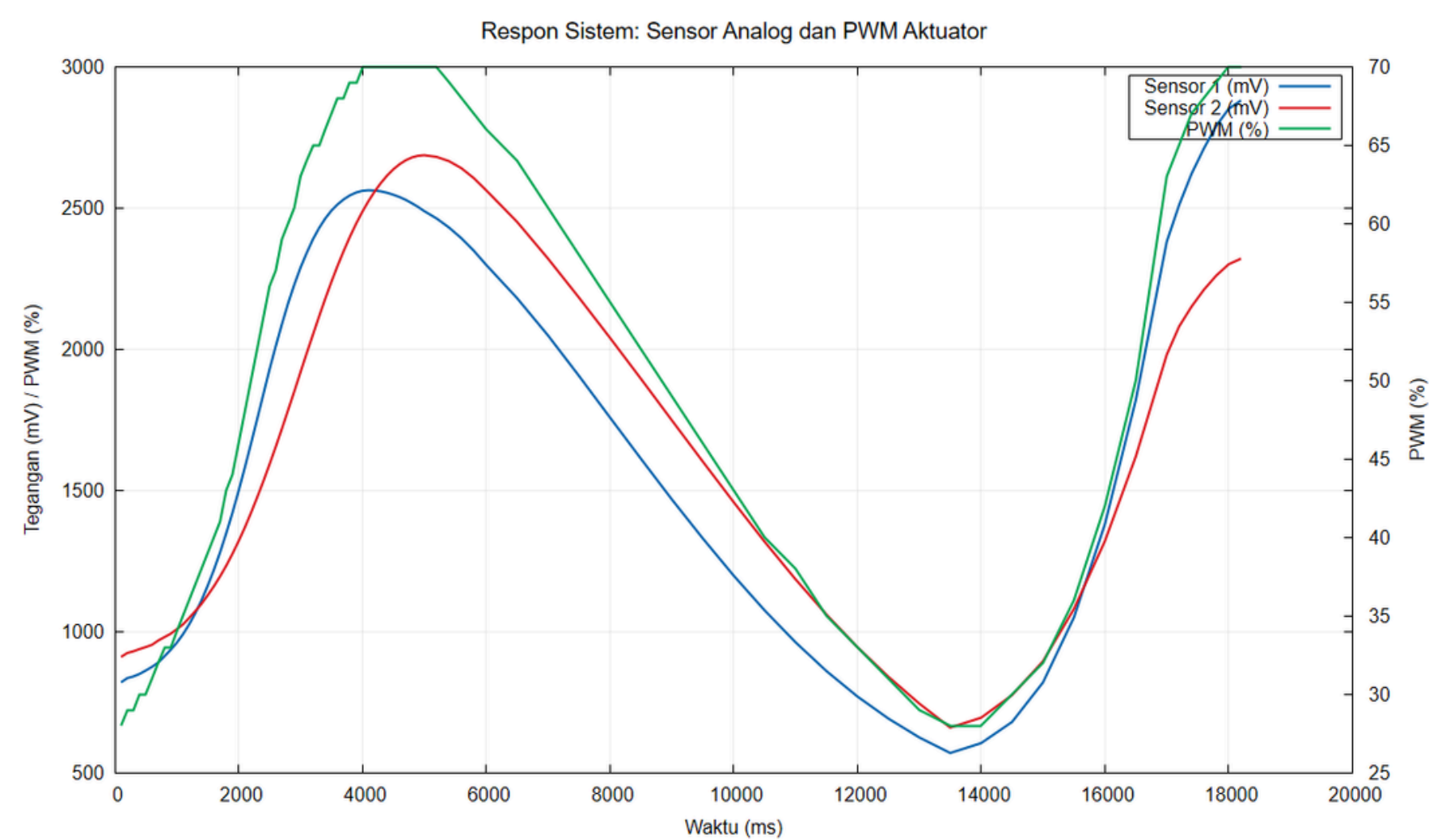


Tabel 3: Rencana uji fungsional dan kriteria lulus

ID	FW	Prosedur	Kriteria Lulus
T1	FW3	Play simulasi; putar POT1	s1_mv di UART berubah monoton
T2	FW8	Observasi CSV saat goyang POT	Sinyal terfilter (MA $N=8$) lebih halus
T3	FW3	Osiloskop PA8	PWM ≈ 10 kHz; duty mengikuti sensor
T4	FW10	Bandingkan interval timestamp CSV	10 ms (dinamis) vs 100 ms (stabil)
T5	FW15	POT di tengah (1,65 V)	state=0, LED hijau PD12 ON
T6	FW15	POT ke 0 V, 1–2 siklus	state=1, LED kuning PD13 ON
T7	FW15	POT max ≥ 3 siklus	state=2, reset WWDG, LED merah PD14 ON
T8	FW11	Virtual Terminal 115200	Baris CSV valid, System Start muncul
T9	FW11	LCD I2C PCF8574	Status sistem tampil di LCD LM016L
T10	FW6	Saat FAULT, pantau task Safety	State FAULT konsisten, task tetap berjalan

Set uji menysar verifikasi tiap komponen sintetis. Format log UART: timestamp_ms,s1_mv,s2_mv,pwm_pct,state.

Data Sampel & Grafik Responst



Thank You